CSE558 · DATA SCIENCE — LECTURE 11

From Polynomial to Exponential

Supratim · IIIT-Delhi · Monsoon 2026

WHERE WE STOPPED

RECALL

Two questions from Lecture 10

Answer out loud — we're about to sharpen both.

Q1 State Markov's inequality: for $X \geq 0$ and $a > 0$, $P(X \geq a) \leq$?

$\frac{E(X)}{a}$

How many chairs does the branch need?

A bank's waiting area has a random number of customers X seated at any moment, with E[X] = 8. There are 15 chairs.

(a) Bound the probability that more than 15 customers are present at once.
(b) If the manager wants the above probability at most 5%. How many customers are needed?

a) $Pr(X \geq 15) \leq \frac{E[X]}{15} = \frac{8}{15}$

$Pr(X \geq a) \leq \frac{E[X]}{a} = 0.05$

What the mean can't see: spread

Q Our coin: 100 tosses, 20% heads, S = #heads. What did Markov bound P(S ≥ 50) at?

indepnt

$E[S] = 0.2 \times 100 = 20$

$\leq \frac{20}{50} = 0.4$

$S = \sum_{i=1}^{100} X_i$

$X_i = \begin{cases} 1 & \text{if heads w.p. } 0.2 \\ 0 & \text{else w.p. } 0.8 \end{cases}$

$Var(X_i) = E[(X_i - E[X_i])^2]$

$Var(X_i) = E[X^2] - (E[X])^2 = 0.2 - 0.04 = 0.16$

$Var(S) = Var\left(\sum_{i=1}^{100} X_i\right) = \sum_{i=1}^{100} Var(X_i) = 16$

Var[S] for our 100 independent tosses of the 20%-heads coin?

CHEBYSHEV'S INEQUALITY

Add the variance, square the payoff

For any random variable X with finite variance, and any a > 0:

$$\mathrm{P}\big(\,|X - \mathrm{E}[X]| \geq a\,\big) \;\leq\; \frac{\mathrm{Var}[X]}{a^2}$$

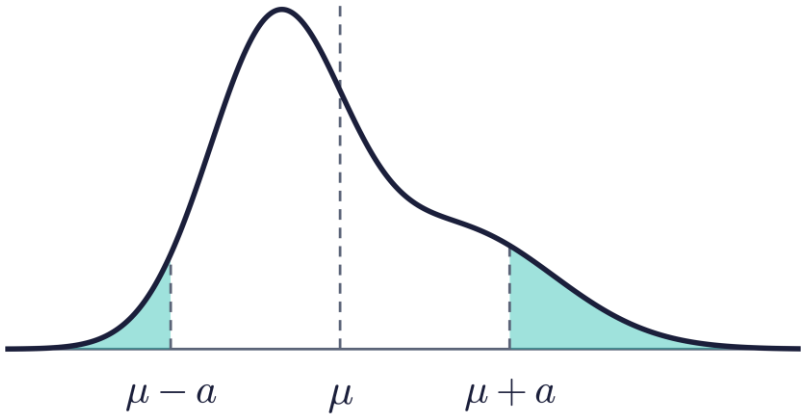
Equivalently, writing a = kσ:

$$\mathrm{P}\big(\,|X - \mu| \geq k\sigma\,\big) \;\leq\; \frac{1}{k^2}$$

1 **A new random variable**
Let Y = (X − E[X])². Then Y ≥ 0 and E[Y] = Var[X].

2 **Same event, rewritten**
|X − E[X]| ≥ a exactly when Y ≥ a².

3 **Apply Markov to Y**
P(Y ≥ a²) ≤ E[Y] / a² = Var[X] / a². ■



Any shape: the shaded tails together hold at most Var[X]/a².

QUICK VOTE

How much data can sit 3σ from the mean?

A popular outlier rule flags every point at least 3 standard deviations from the mean. For Normal data that is about 0.3% of points. For an arbitrary dataset (any shape at all), what is the largest fraction that can be that far out?

A 0.3%

B 5%

C 11%

D 33%

$$Pr(|X - E[X]| \geq 3\sigma) \leq \frac{\sigma^2}{9\sigma^2} = \frac{1}{9} \approx 11\%$$

SAME COIN, SHARPER TOOL

Chebyshev on the same 100 tosses

Var[S] = 16. What is the probability of having 50 or more heads?

$$Pr(S \geq 50)$$

$$\leq Pr(|S - 20| \geq 30) \leq \frac{16}{900} \leq 0.017$$

THE SCOREBOARD

Method	Knows	Upper bound on P(S ≥ 50)
Markov	the mean	0.4
Chebyshev	mean + variance	**0.018**
Chernoff	every moment	?
Exact	the full distribution	?

$$Pr(S \geq 50) < 0.018 < 0.4$$

$$Pr(S < 50)$$

YOUR TURN

Will she get the job?

4,000 applicants take an aptitude test; the top 300 get offers. The scores have mean 56 and standard deviation 6 — nothing is known about the shape of their distribution. Your friend scored 80. Should she be optimistic? Justify.

$P_r(X \geq 81) \leq \frac{56}{81}$, $E(\#\text{people} \geq 81) \leq \frac{56}{81} \times 4000 = 2.7K$

$P_r(X \geq 80) \leq P_r(|X - 56| \geq 24) \leq \frac{36}{24^2} = \frac{1}{16}$

$\frac{81 \times 300}{4000} = 26$

$\rightarrow E(\#\text{people} \geq 80 \text{ or } \leq 32) \leq \frac{4000}{16} = 250$

Is there really an attack?

A server's failed-login count per minute, X, averages E[X] = 4 during normal traffic. Security policy calls a minute an ‘attack minute’ once X ≥ 40.

(a) Using only Markov's inequality, bound P(attack in a given minute).

(b) During an attack minute the intrusion alarm fires with probability 0.9; during a normal minute it fires falsely with probability 0.01. The alarm just fired. Using (a)'s bound as an upper bound on the prior P(attack), conclude it this is really an attack?

THE ROADMAP, REVISITED

One tool left

More information about X buys a sharper bound. Today we reach the tool that uses everything — every moment of X at once.

MORE INFORMATION ABOUT X → SHARPER BOUND

One function, every moment

The m.g.f. of a random variable X is: $M_X(t) = E[e^{tX}]$

$M_X(t) = E[e^{tX}] = E\left[1 + \sum_{i=1}^{\infty} \frac{(tx)^i}{i!}\right] = 1 + \frac{tE[X]}{1} + \frac{t^2 E[X^2]}{2!} + \cdots$

$e^a = 1 + a + \frac{a^2}{2!} + \cdots$

$\left.\frac{\partial M_X(t)}{\partial t}\right|_{t=0} = 0 + E[X] + \frac{2tE[X^2]}{2!} + \cdots$

$\left.\frac{dM_X(t)}{dt}\right|_{t=0} = E(X)$

$\left.\frac{d^2 M_X(t)}{dt^2}\right|_{t=0} = E(X^2)$

$\left.\frac{d^k M_X(t)}{dt^k}\right|_{t=0} = E[X^k]$

For independent X and Y, the m.g.f. of a sum is a product:

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

Markov, aimed at a different point

For any random variable X and any $a \in \mathbb{R}$:

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}} \quad \text{and} \quad P(X \leq a) \leq \min_{t<0} \frac{E[e^{tX}]}{e^{ta}}$$

1 **For any t > 0, exponentiate**

X ≥ a exactly when $e^{tX} \geq e^{ta}$.

2 **Apply Markov**

$$Y = e^{tX}, \quad Pr(X > a) = Pr(Y \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

3 **This holds for every t > 0**

So pick whichever t makes the bound tightest — hence the min over t.

Building for 100 tosses

Our running example: S = number of heads in 100 independent tosses of a 0.2-heads coin. We want P(S ≥ 50).

A single toss:

$$X_1 = \begin{cases} 1 & \text{if head} \\ 0 & \text{else} \end{cases} \qquad M_{X_1}(t) = E[e^{tX_1}] = 0.2e^t + (1-0.2)e^0 = 1 + 0.2(e^t - 1)$$

All 100, independent — multiply:

$$S = \sum_{i=1}^{100} X_i \qquad M_S(t) = \prod_{i=1}^{100} e^{0.2(e^t-1)} = e^{\sum_{i=1}^{100} 0.2(e^t-1)} = e^{20(e^t-1)} \le e^{0.2(e^t-1)}$$

Apply Markov, with a = 50, then substitute:

$$\Pr(S \ge 50) = \Pr(e^{tS} \ge e^{t50}) \le \frac{e^{20(e^t-1)}}{e^{t\,50}} = \left(\frac{e^{e^t-1}}{e^{25t}}\right)^{20}$$

$$t = \ln(2.5)$$

$$e^t = 2.5$$

$$\left(\frac{e^{1.5}}{(2.5)^2}\right)^{20} =$$

$$e^a = 1 + a + \frac{a^2}{2!} + \frac{a^3}{3!} + \cdots$$

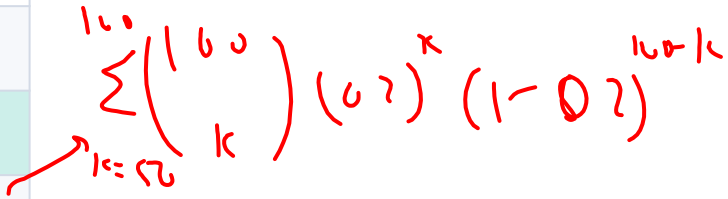
THE REVEAL

1.35 in ten million

Setting t = ln(2.5):

$$\mathrm{P}(S \geq 50) \;\leq\; \left(\frac{e^{1.5}}{2.5^{2.5}}\right)^{20} \;\approx\; 1.35 \times 10^{-7}$$

Method	Knows	Upper bound on P(S ≥ 50)
Markov	the mean	0.4
Chebyshev	mean + variance	0.018
Chernoff	every moment	**1.35 × 10⁻⁷**
Exact	the full distribution	?



The Chernoff bound on a sum of Poisson trials

Let $X_1, \ldots, X_n$ be independent 0/1 trials with $P(X_i=1) = p_i$ (not necessarily equal), and $X = \Sigma X_i$.

Mean: $$\mu = \mathrm{E}[X] = \sum_{i=1}^{n} p_i$$

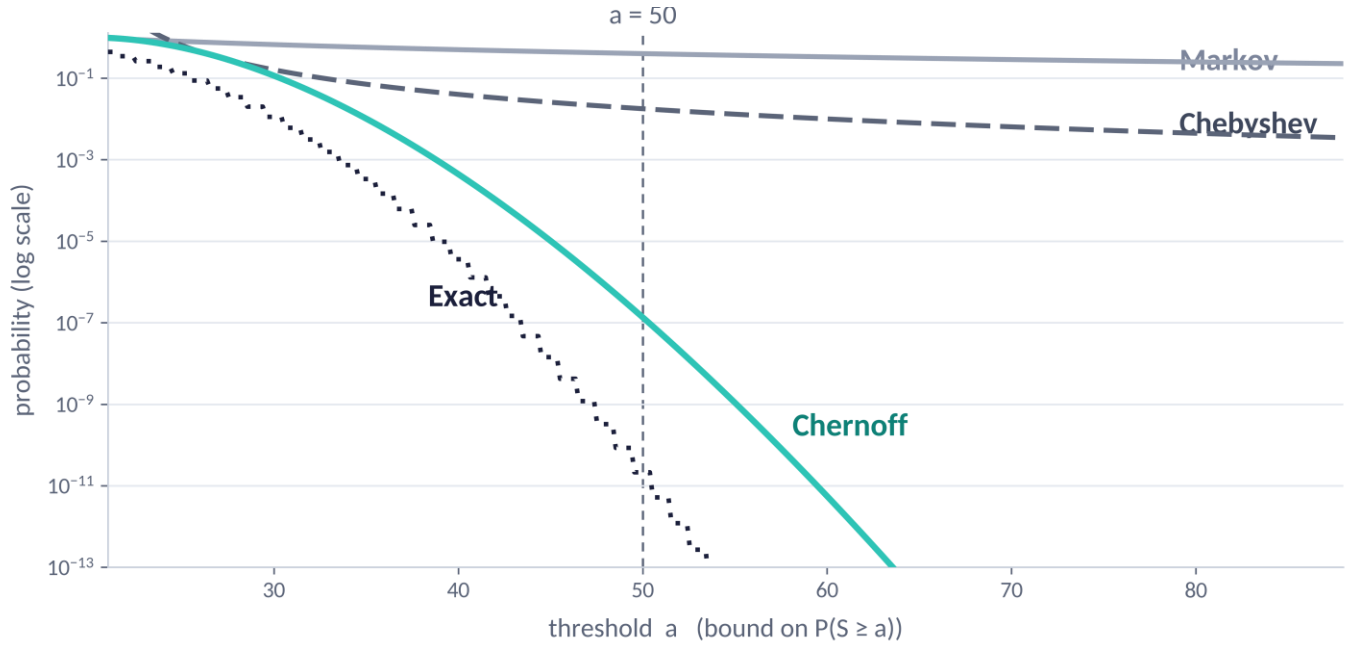
YOUR TURN

PAIR UP

Is a server about to overload?

1,000 requests arrive and each, independently, is routed to one of 10 servers uniformly at random. Watch one specific server; let X be its load. What is the probability that the server gets a load more than 149 requests.

How unlikely was 50 heads, really?



EXIT TICKET

Is the judge under-accepting?

An online judge processes 500 independent submissions an hour. Historically each is Accepted with probability 0.3. Bound the probability that at most 90 submissions are accepted this hour.

STUDY MATERIAL

Reference

Probability and Computing: Chapter 3.2,4.1,4.2 by Mitzenmacher & Upfal

Slides